

Analysis of the Integrated Assessment Model (IAM) Scenario

The provided data presents a comprehensive analysis of an IAM scenario, focusing on various energy-related metrics and their projected changes from 2025 to 2070. This analysis aims to break down the key trends and insights from the data.

Resource and Primary Energy Trends

- **Resource:** The total volume of resources is expected to increase significantly, from 414,490.905 in 2025 to 2,177,408.112 by 2070. This represents a relative increase of 425.32% and a compound annual growth rate (CAGR) of 3.76%.
- **Primary Energy (substitution method):** There's a notable increase in primary energy, rising from 71,570.834 in 2025 to 301,159.333 by 2070. This marks a 320.78% relative increase and a 3.24% CAGR.

Emissions and Cost Analysis

- **Emissions:** Emissions are projected to rise from 8,252.872 in 2025 to 37,932.067 by 2070, indicating a 359.62% relative increase and a 3.45% CAGR.
- **Cost:** The total cost is expected to increase from 113.2 in 2025 to 473.887 by 2070, showing a 318.63% relative increase and a 3.23% CAGR.

Final Energy and Sectoral Analysis

- **Final Energy:** Final energy consumption is anticipated to grow from 63,108.537 in 2025 to 282,076.402 by 2070, representing a 346.97% relative increase and a 3.38% CAGR.
- **Sectoral Market Shares:** By 2070, the market shares in final energy consumption are expected to be dominated by Industry (22.99%), followed by Residential and Commercial (16.48%), and Transportation (8.9%).

Capacity and Investment Trends

- **Cumulative Capacity:** There's a significant increase in cumulative capacity, from 1,047.321 in 2025 to 20,667.071 by 2070, marking a 1,873.33% relative increase and a 6.85% CAGR.
- **Investment:** Investment in the energy sector is projected to rise from 175.85 in 2025 to 843.647 by 2070, indicating a 379.75% relative increase and a 3.55% CAGR.

Carbon Sequestration and Other Metrics

- **Carbon Sequestration:** Carbon sequestration efforts are expected to vary, with a peak in 2045 at 484.453, then fluctuating and ending at 300.477 by 2070.

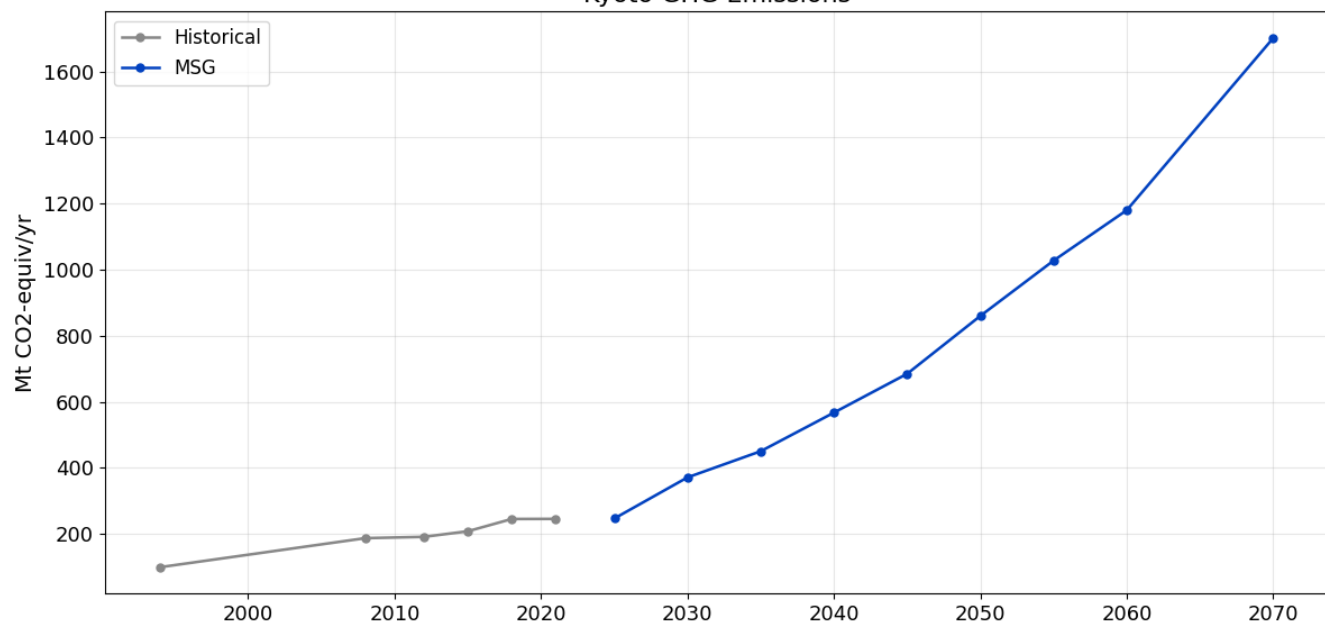
Summary

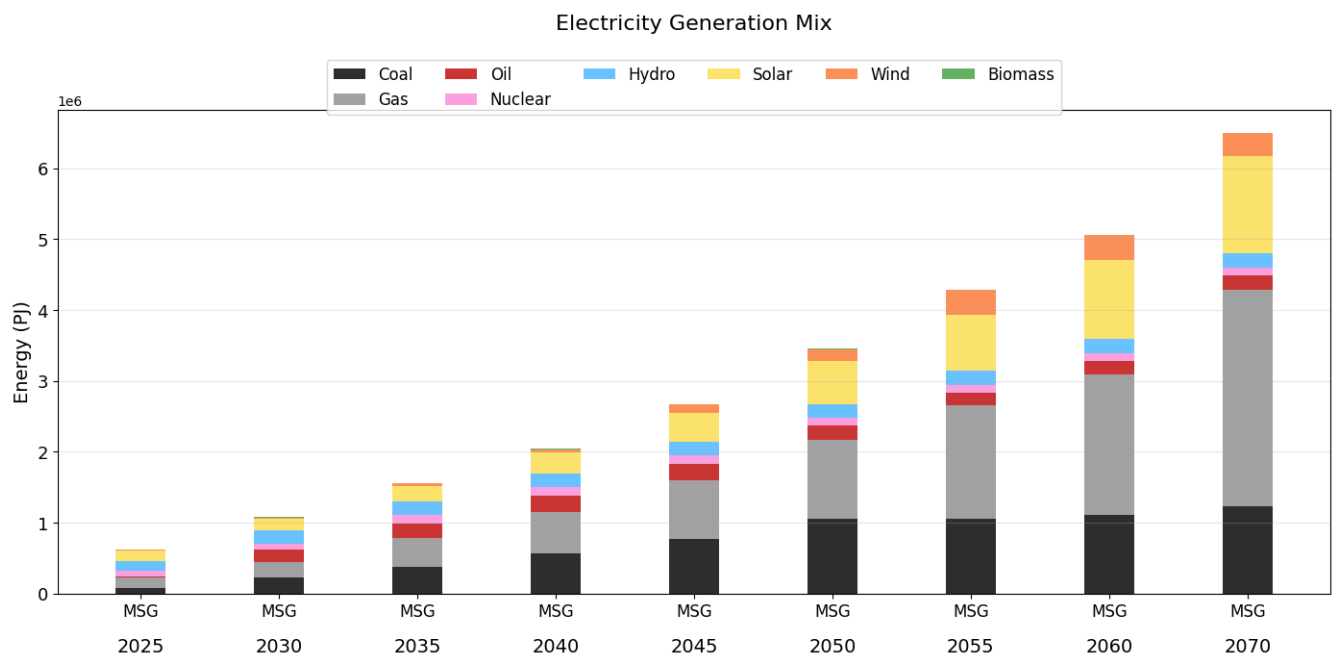
Key Points:

- Significant growth in resource utilization, primary energy consumption, and final energy demand.
- Emissions are expected to rise substantially, alongside increases in cost and investment in the energy sector.
- Sectoral shifts in final energy consumption and a notable increase in cumulative capacity and investments.
- Variability in carbon sequestration efforts over the years.

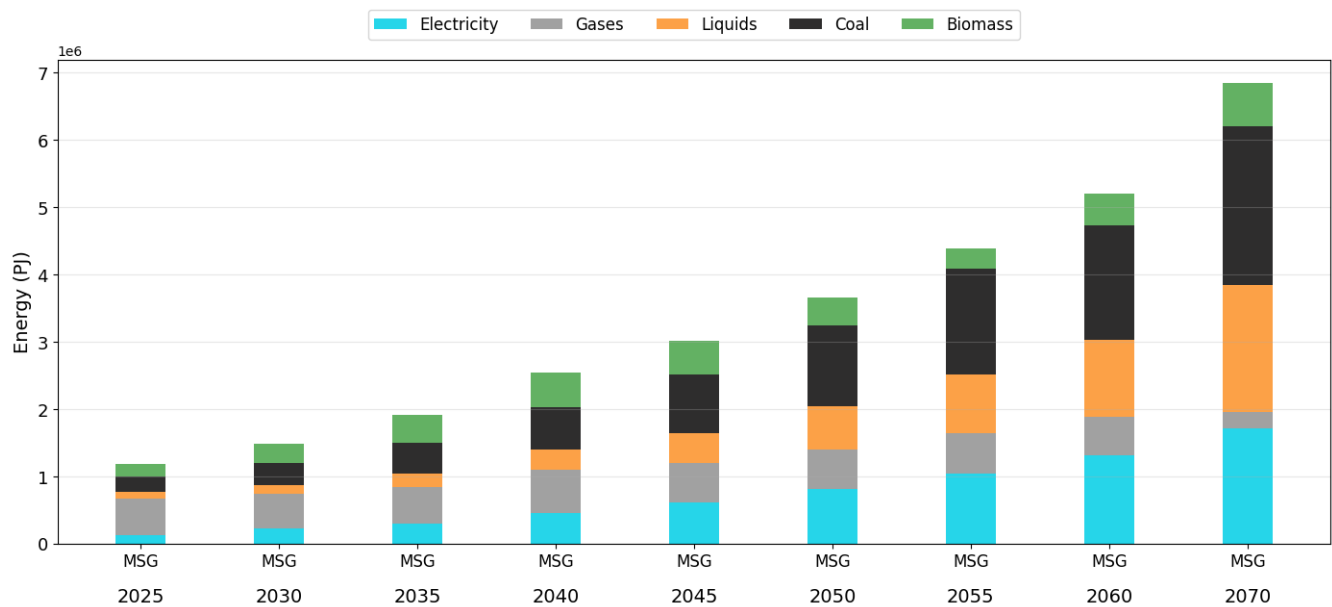
This analysis provides a broad overview of the trends and insights from the IAM scenario, highlighting areas of growth, shifts in sectoral demands, and the challenges posed by increasing emissions.

Kyoto GHG Emissions

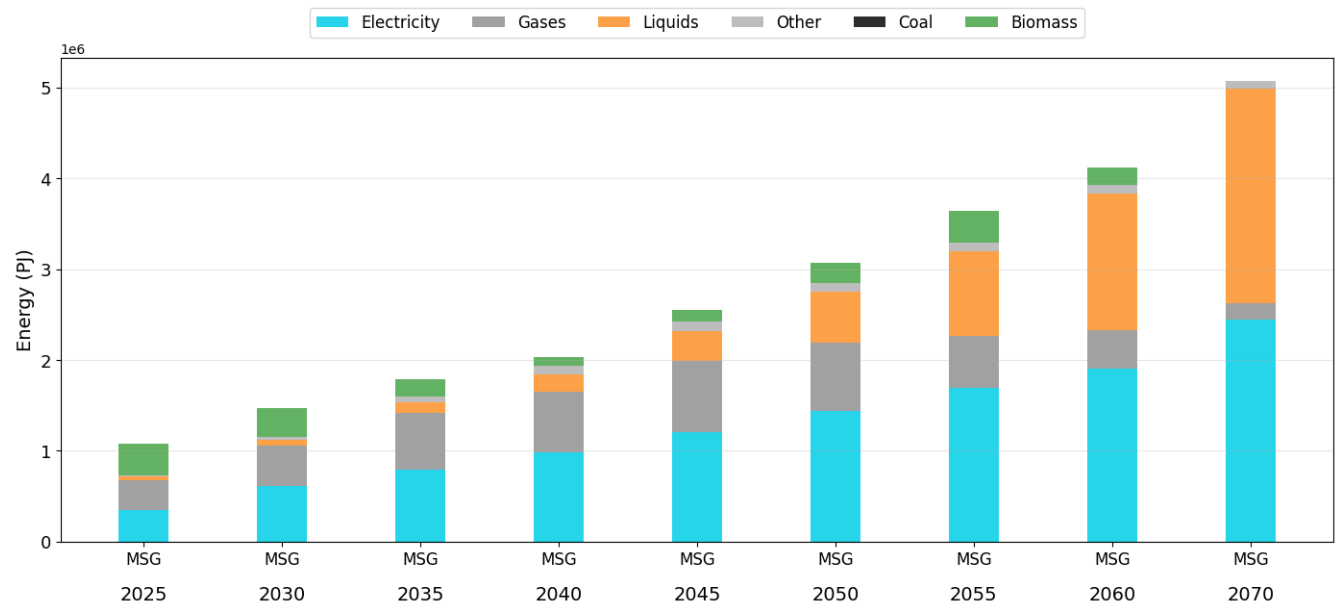


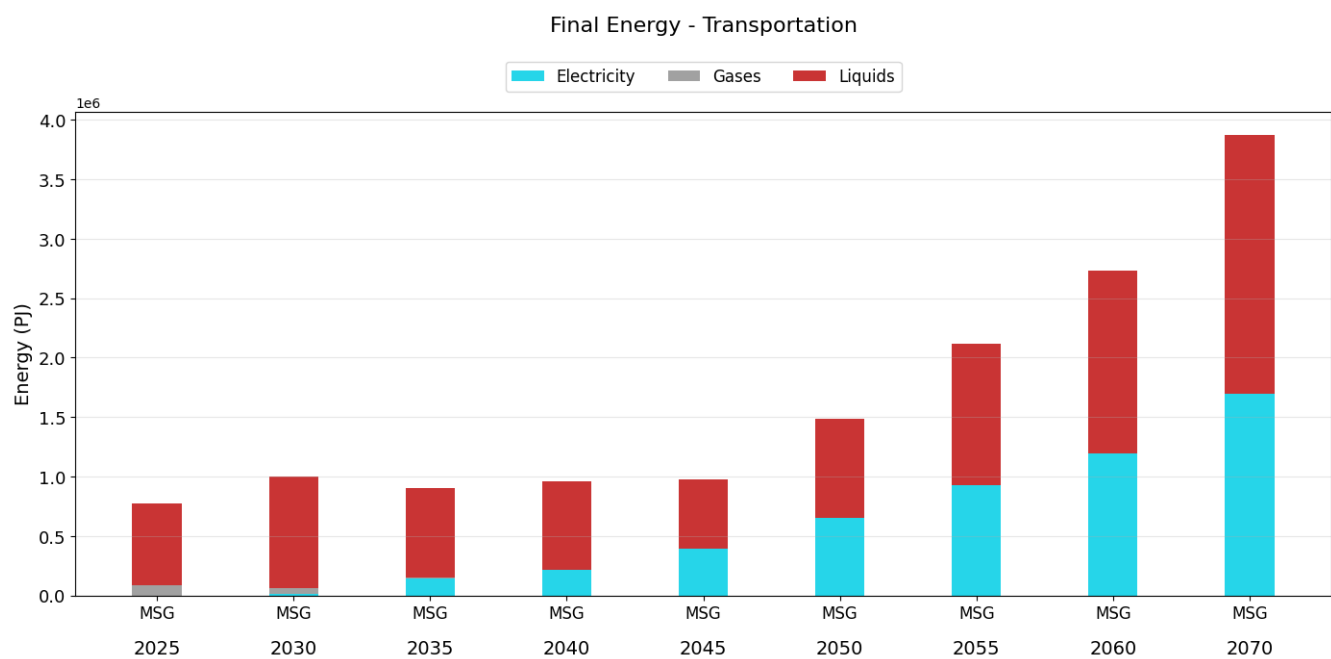


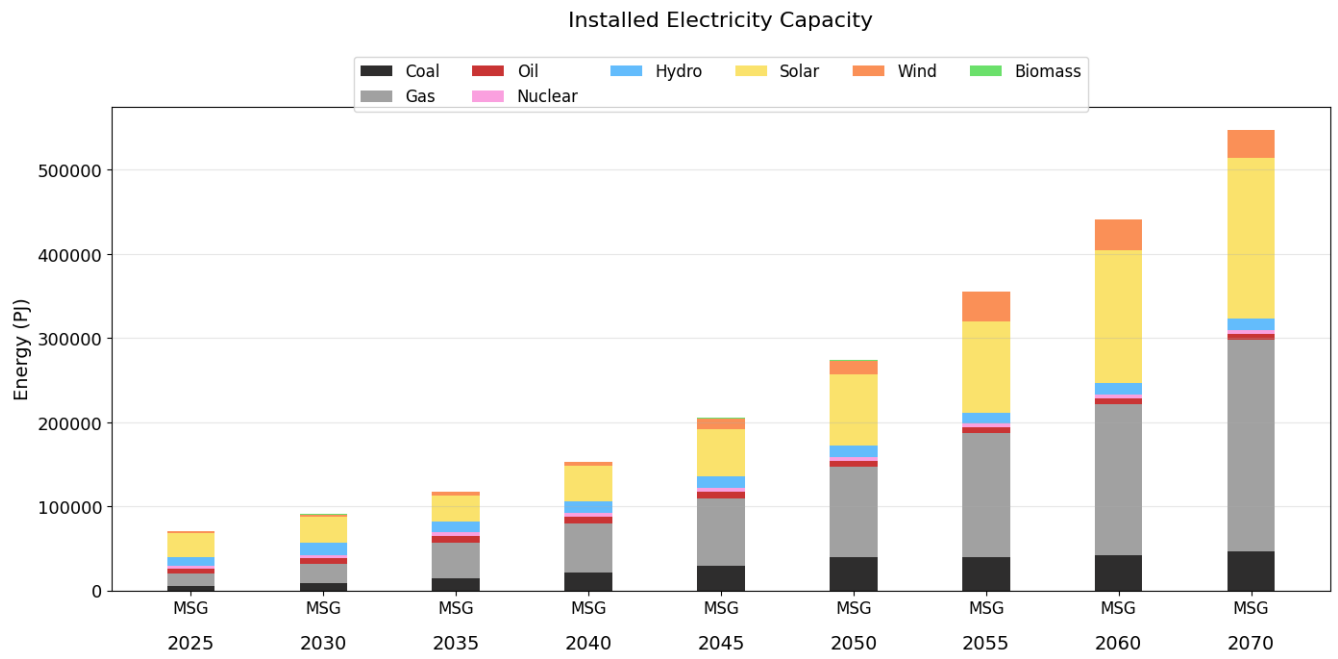
Final Energy - Industry



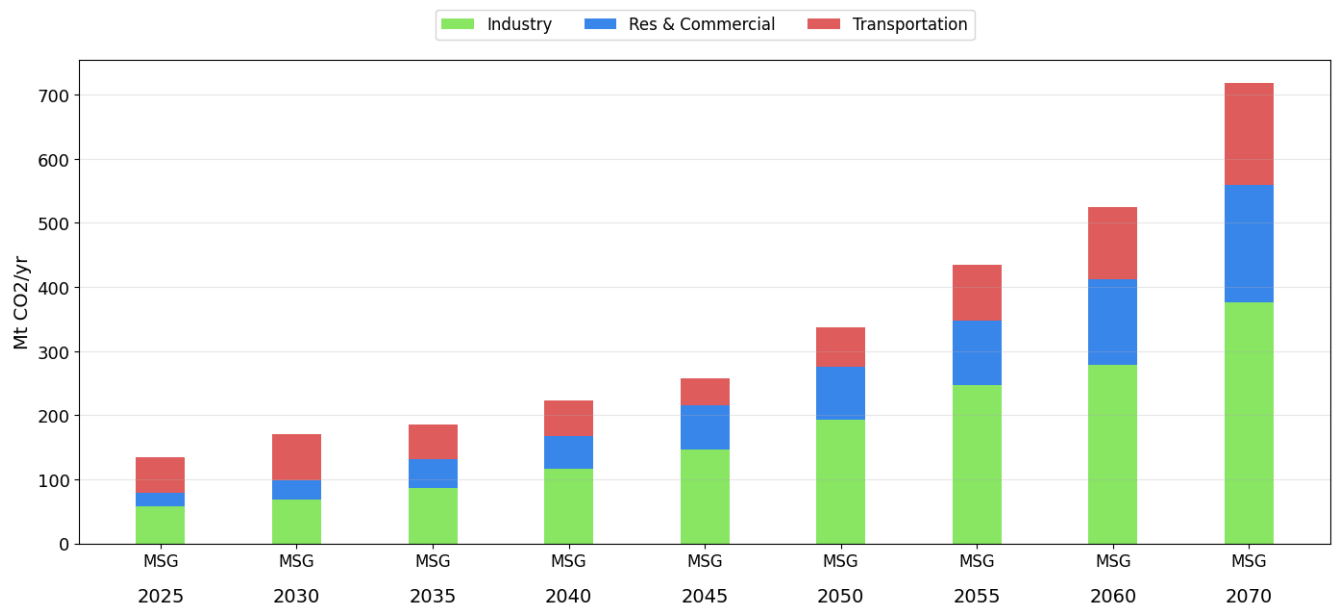
Final Energy - Residential & Commercial



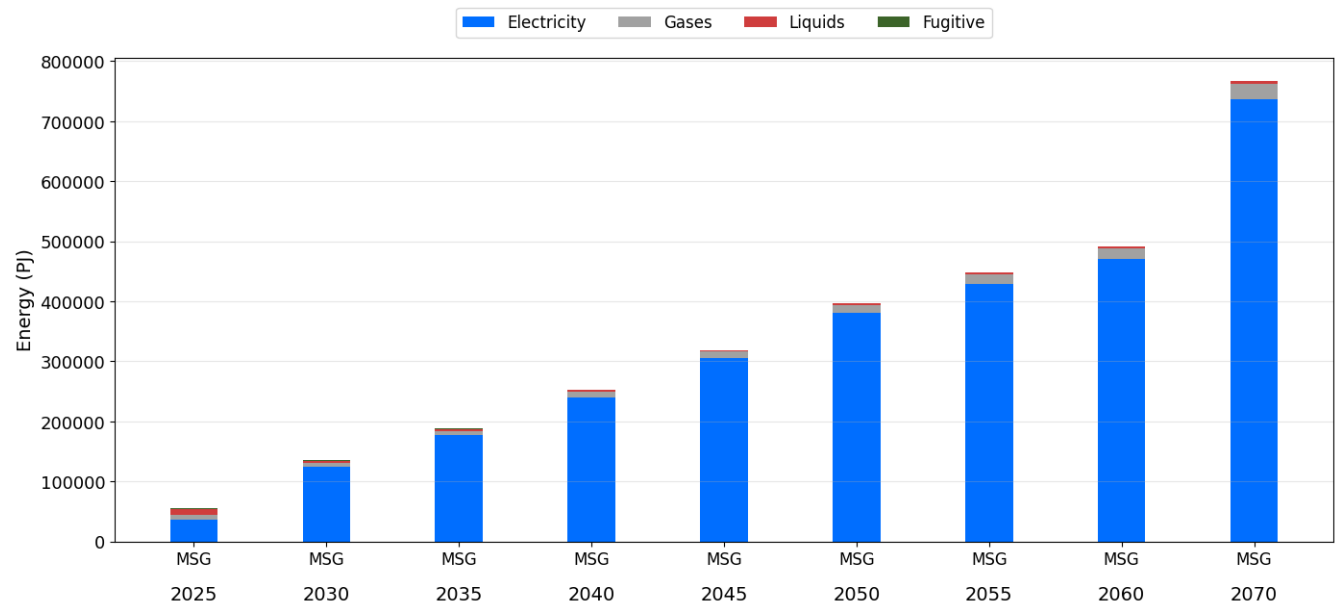


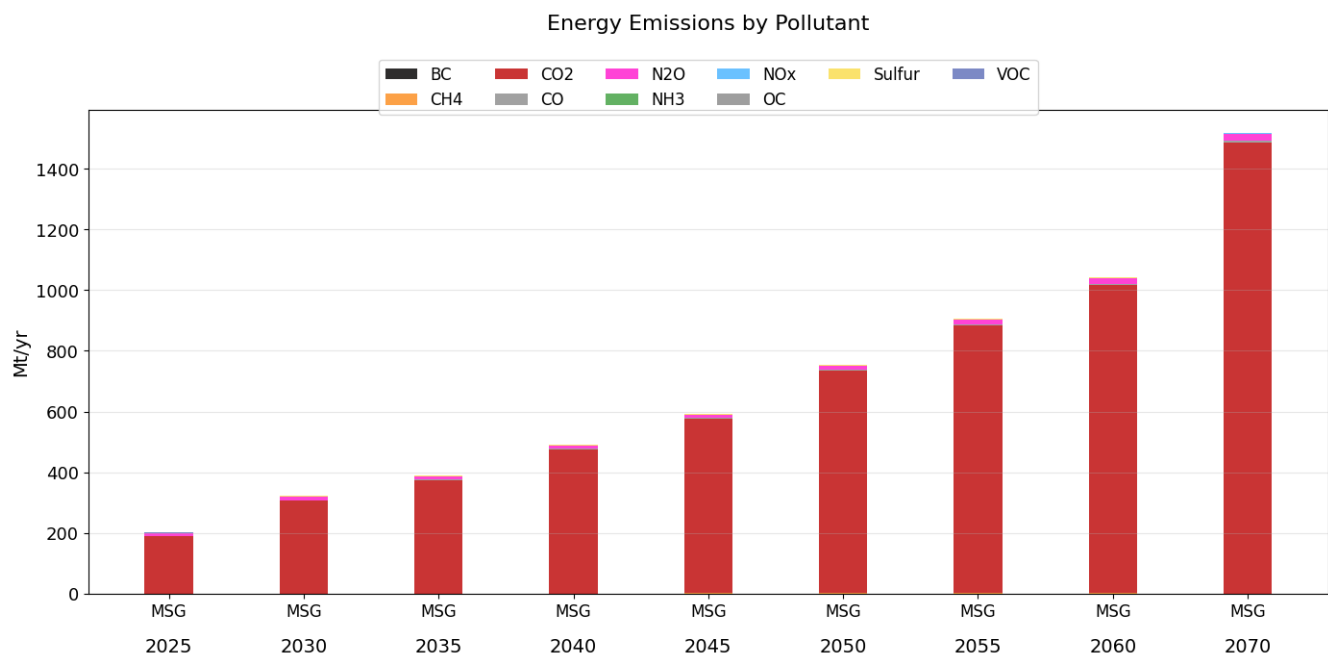


CO2 Emissions by Demand Sector

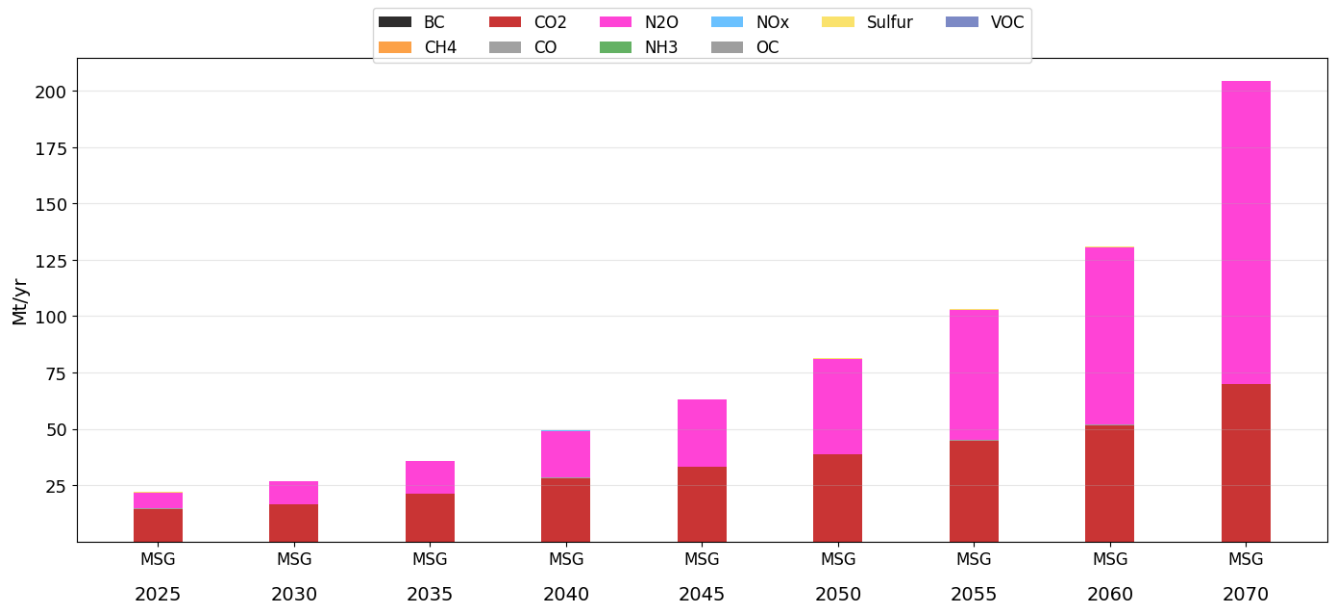


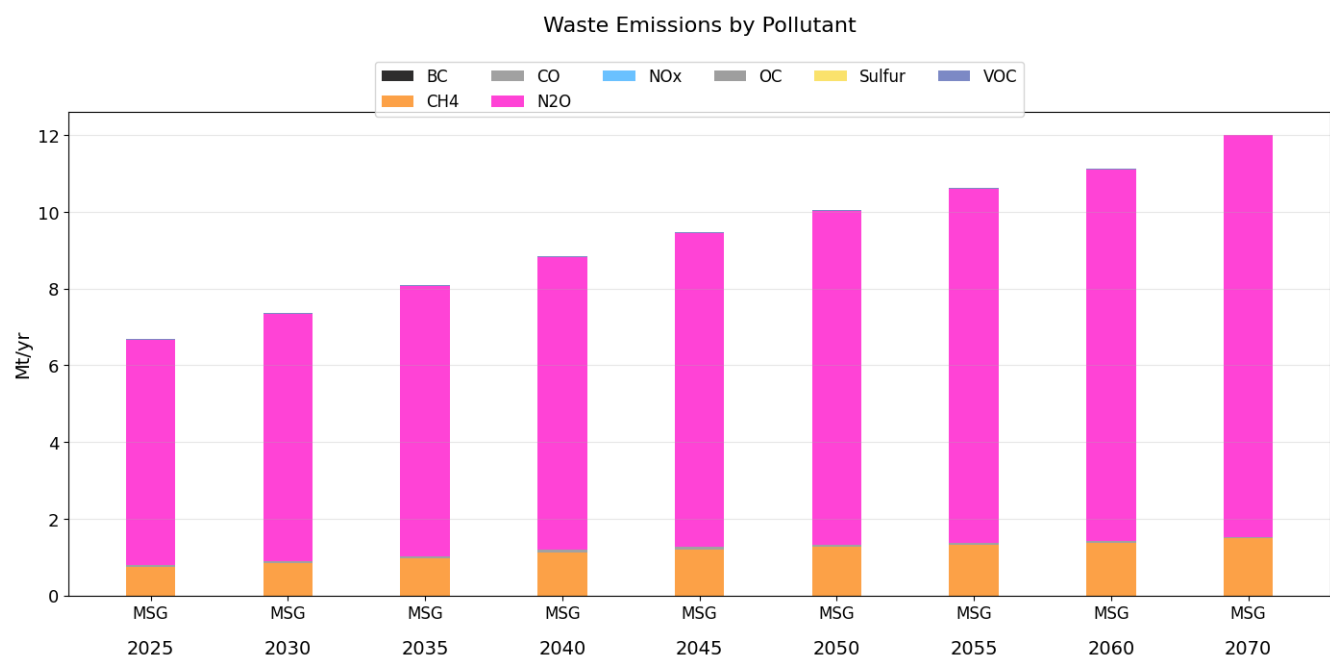
CO2 Emissions by Energy Supply

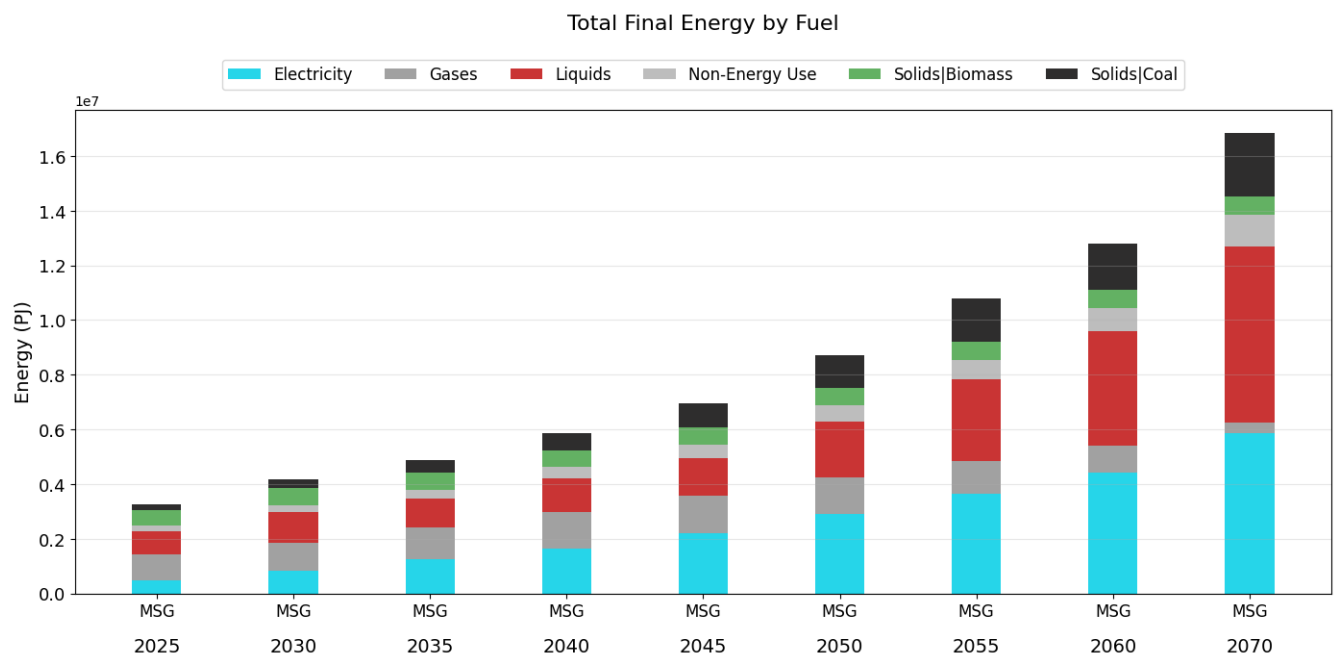




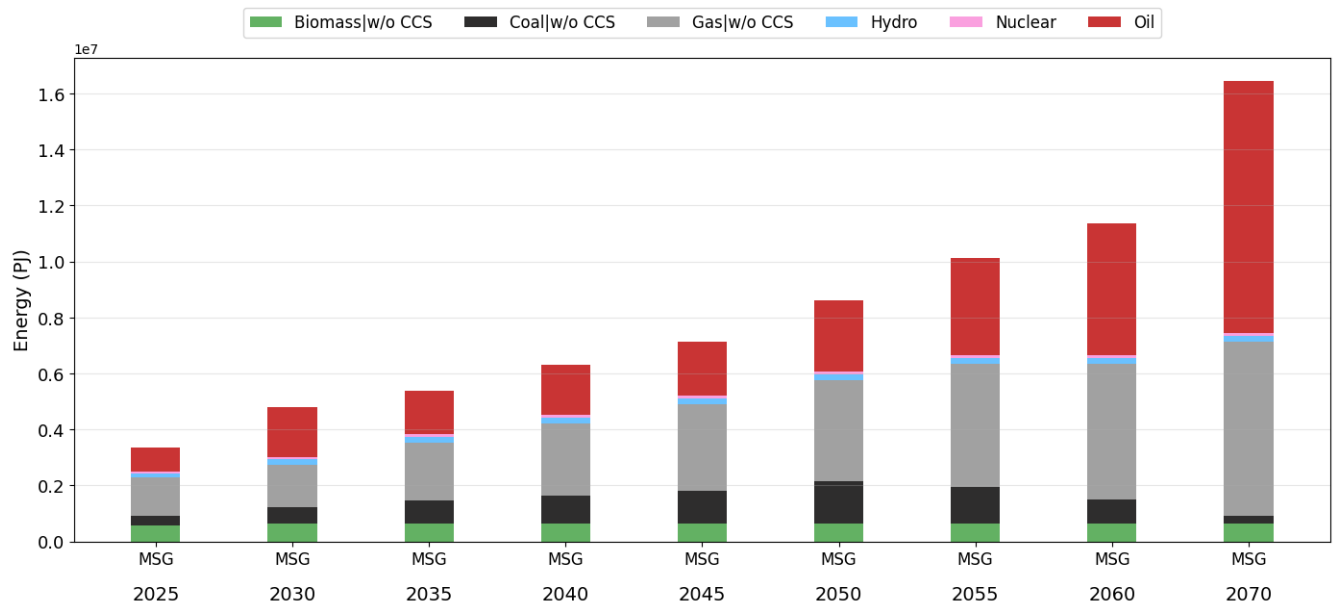
Industrial Process Emissions by Pollutant



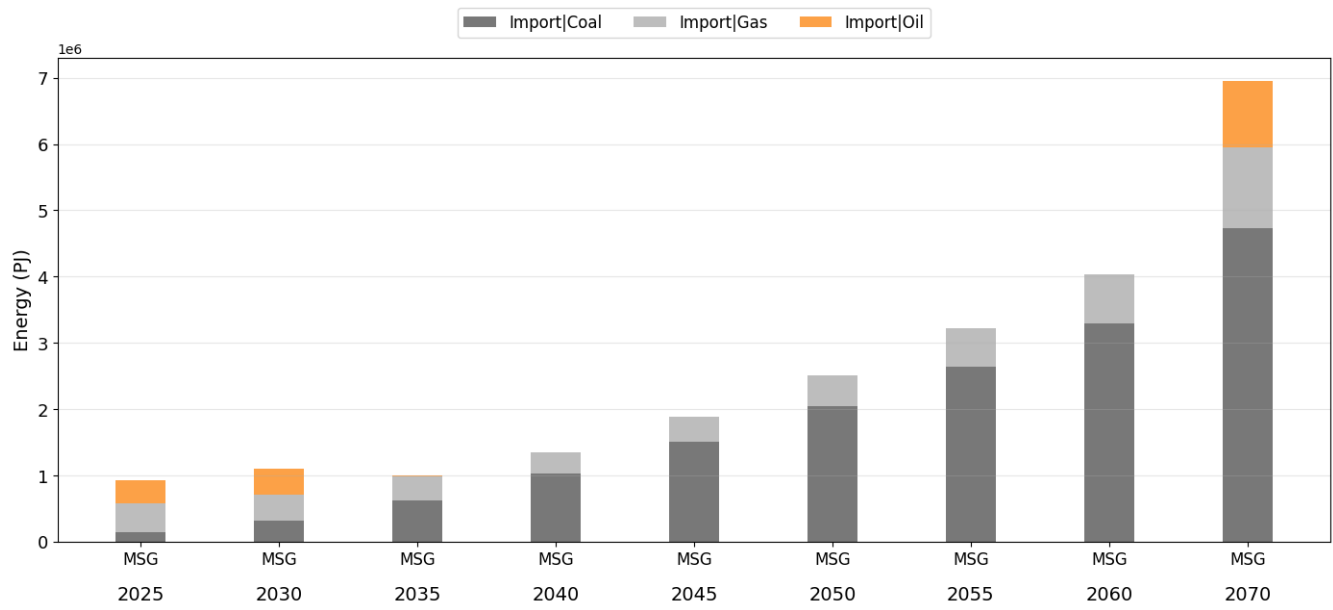




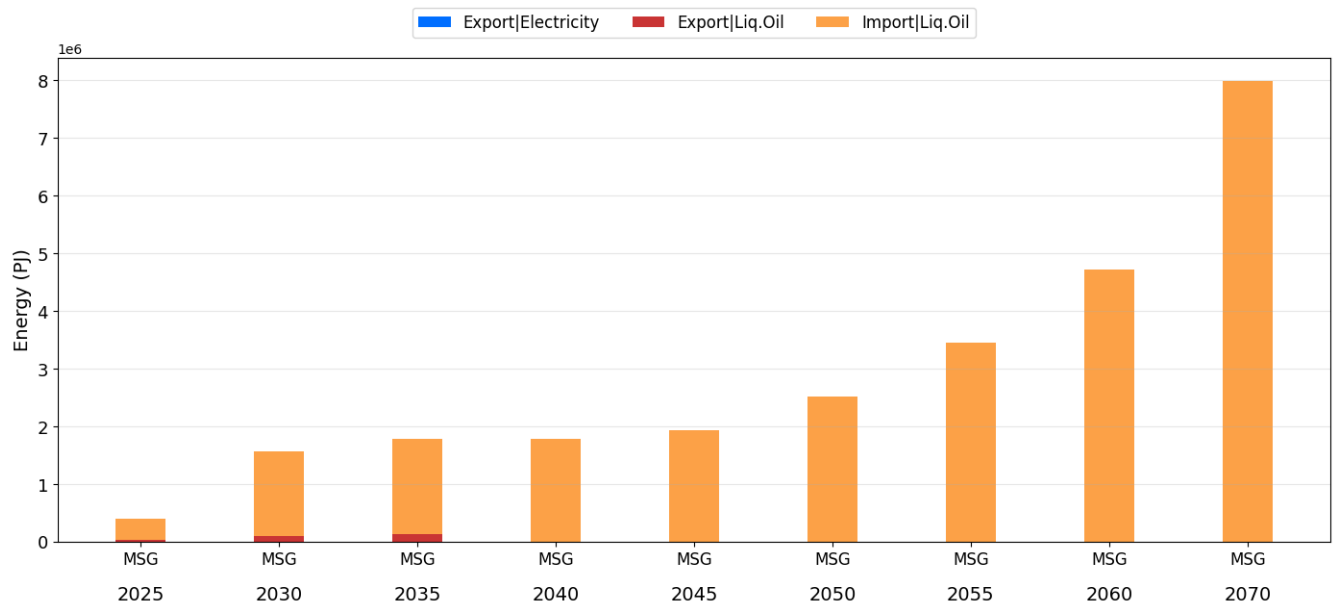
Primary Energy Mix



Trade - Primary Energy Volumes



Trade - Secondary Energy Volumes



Resource Extraction

